

Shift-Aware Candidate Selection for Stochastic Pairwise MIS in Real-Time ReSTIR: Realization, Limits, and Diagnostics

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[Teaser figure slot] Planned: equal-time comparison on a cluttered scene under camera motion - pair-mixture reuse / SPMIS with unconditional search / SPMIS with the tiny-run gate - full frames plus disocclusion and glossy crops, with per-crop MAPE. To be rendered by the final experiment protocol (Release build).

Abstract

Spatial reuse in ReSTIR draws a handful of neighbor pixels and merges their reservoirs with resampling MIS weights. Hedstrom et al. [2026] showed that the neighbors themselves may be chosen *based on their samples* without introducing bias, provided the selection probabilities $P(i)$ are compensated inside stochastic resampling MIS weights; their SPMIS estimator selects proportionally to the *unshifted* source-pixel contribution $c_i \cdot p(X_i) \cdot W_i$, and names selection proportional to the true post-shift resampling weight $m_i(Y_i) \cdot p(Y_i) \cdot W_i \cdot |\partial T / \partial X_i|$ as an open problem: evaluating it needs the very shift mappings the method tries to avoid.

We realize that selection density and evaluate it honestly. The missing factor - the shift survival ratio $\tilde{p}(T(X)) \cdot |\partial T / \partial X| / p(X)$ - is observed for free for every shift a ReSTIR renderer already evaluates; we learn it online as a per-pixel exponential moving average that reprojects with the temporal history and resets on disocclusion. Because the stochastic MIS weights compensate ANY positive selection distribution, a mispredicted score can only cost variance, never correctness. We further prove a first-moment lemma - the stochastic-MIS estimator remains unbiased under any candidate-drawing scheme whose per-draw marginals match $P(i)$, not just i.i.d. multinomial sampling - which licenses antithetic stratified draws on the selection CDF and history-dependent selection policies.

The empirical verdict is a finding in itself and splits by REGIME. On contribution-DENSE pools (Cornell-class scenes, where $\sim 99.7\%$ of a block run's entries carry nonzero selection mass), selection is SATURATED. The sound (bounded-ratio) form of survival-guided selection changes noise by at most $\pm 2\%$ at $\tilde{N} \in \{3, 8\}$ on clutter, glossy, and disocclusion stress tests, and the naive form - selecting by the realized post-shift CONTRIBUTION - is actively harmful: its heavy-tailed scores concentrate P until the $1/(\tilde{N}P)$ compensation sprays fireflies (a failure mode we attribute with a defensive-mixture dose-response). Two byproducts outweigh the intended contribution. First, SPMIS's unconditional neighborhood search

is a measured net harm on this scene class - probing replaces close partners with distant ones whose shifts fail more, monotonically in probe count and radius, and even wide fresh disocclusion strips reuse better from their own same-frame runs; gating the search to near-empty runs keeps its rescue role at zero steady-state tax. Second, a dual-direction pair diagnostic that runs BOTH shift directions of the same (canonical, partner) pair bounds the shift machinery's forward/backward asymmetry at $1e-7$ locally - refuting asymmetry as the cause of the chain's small positive soak residual - and identifies the true mechanism: direction-symmetric FALSE-NEGATIVE shift failures feeding the stochastic weight functional, whose magnitude tracks the measured failure rate.

On contribution-SPARSE pools the picture inverts. In a purpose-built scene whose emitter sits behind a two-bend chicane - every visible surface at least two bounces from the light, next-event estimation fully occluded, only $\sim 1\%$ of run entries carrying selection mass and $\sim 44\%$ of pixels sitting in runs with none at all - contribution-proportional selection is the difference between unusable and usable: the stratified pair-mixture baseline is WORSE than no spatial reuse at all (its rank picks feed dead neighbors into the mixture), while SPMIS selection cuts noise 49-67%. The same scene yields a second impossibility-flavored result: making the neighborhood SEARCH contribution-aware (trigger and probe weights on run selection mass) is irreparably biased (+16% - the estimator's accounting requires dead-run realizations to contribute exactly zero, and a value-dependent pool swap fires precisely there), which reframes SPMIS's value-blind confidence weighting as a correctness constraint and points to always-on union pools with exact per-draw compensation as the only unbiased rescue.

Building on the cheap large-kernel machinery, we further demonstrate TEMPORAL-FREE ReSTIR: batching several stochastic candidates through one shift-pipeline sweep makes $\tilde{N} = 16-24$ candidates affordable, at which point spatial-only reuse MATCHES the per-frame quality of full temporal-plus-spatial ReSTIR - with uniform noise instead of a 5x-noisier disocclusion band, correct energy under motion instead of

transient history-invalidation darkening, and no temporal lag or correlation trails, at roughly twice the (unoptimized) reuse cost.

1 Introduction

ReSTIR amortizes path sampling across pixels and frames by resampling: each pixel keeps one reservoir sample and improves it by merging candidates borrowed from spatiotemporal neighbors under generalized resampled importance sampling (GRIS). WHICH neighbors to borrow from has remained surprisingly primitive: production implementations still draw a few pixels uniformly from a disk and reject on G-buffer differences, essentially unchanged since Bitterli et al. [2020]. Uniform selection wastes shifts in exactly the situations where reuse matters most - when few nearby reservoirs hold contributing samples (disocclusions, sparse initial sampling), or when many hold bright samples whose shift into the receiver dies (shadow edges, glossy transport, geometric discontinuities).

Stochastic pairwise MIS (SPMIS) [Hedstrom et al. 2026] is the enabling theory for doing better: reuse mathematically from a large pool of M inputs, evaluate only $\tilde{N}$ of them drawn with probabilities $P(i)$ that MAY depend on the samples, and multiply each drawn candidate's MIS weight by $K(i)/(\tilde{N}P(i))$. Any positive P yields an unbiased estimator; P only steers variance. SPMIS instantiates P with the source-pixel contribution - the sample's brightness where it ALREADY is - which ignores everything about the shift into the receiver. The authors state the ideal selection density (the post-shift resampling weight) and leave its approximation open. Concurrently, Junkins et al. [2026] select neighbors by an analytic G-buffer compatibility score - a shift-quality prior that ignores the samples - and name the combination with SPMIS as future work. Neither line uses the one signal a ReSTIR renderer produces in abundance every frame: the realized outcomes of the shifts it evaluates.

We treat the ideal selection weight as a product of what SPMIS already uses (source contribution) and a shift-survival factor, estimate the latter by online learning from the renderer's own shift outcomes, and report what a careful evaluation actually shows. Our contributions:

- The first realization of **survival-guided selection**: a per-pixel score learned online from realized shift outcomes, reprojected along temporal history and reset at disocclusions, folded into SPMIS's selection weights without extra rays and without bias for any predictor quality - together with the honest empirical verdict that block-local selection is SATURATED (the bounded-ratio form is a wash; the absolute-contribution form fireflies via $1/P$ inflation, with the failure mechanism attributed).
- A **first-moment lemma**: the stochastic resampling MIS weight remains unbiased under any drawing scheme with $E[K(i)] = \tilde{N}P(i)$, covering stratified, antithetic, and history-dependent selection. This licenses **antithetic CDF draws** - stratification and importance selection compose.
- A data-driven **starvation gate** for SPMIS's neighborhood search: unconditional probing is a measured 23-45% steady-state noise tax on block-run pools (the own run survives the probe WRS with probability $\sim 1/(1+\text{probes})$),

and even wide fresh disocclusion strips prefer their own same-frame runs; restricting the search to near-empty runs ($\text{count} \leq 2$) keeps its rescue role at zero measured cost.

- A **dual-direction pair diagnostic** that evaluates both shift directions of the same (canonical, partner) pair each frame, plus an involutive fixed-stride pairing that keeps distant-pair ensembles exchangeable. It bounds the shift machinery's directional asymmetry at $1e-7$ locally and identifies the chain's small positive residual as direction-symmetric false-negative shift failures amplified by the stochastic weight functional - with the failure-rate dose-response to match.

- An implementation on block-local sorted cluster runs with exact $O(\log M)$ CDF inversion (segmented scans in the existing histogram-stratification sort), integrated in a wavefront ReSTIR BDPT renderer; the stochastic chain is CHEAPER than the pair-mixture baseline it replaces, and batching candidates through one shift-pipeline sweep makes $\tilde{N} = 16-24$ affordable (the temporal-free regime).

2 Related Work

ReSTIR and GRIS. Reservoir resampling for direct lighting [Bitterli et al. 2020] generalizes to path tracing through GRIS [Lin et al. 2022], which introduces unbiased contribution weights, shift mappings with Jacobians, confidence weights, and the generalized balance and pairwise MIS heuristics we build on; see the course of Wyman et al. [2023]. Extensions cover indirect gathering, area sampling, subsurface scattering, bidirectional techniques, and volumes; all inherit Bitterli-style uniform neighbor draws with G-buffer rejection.

Neighbor selection. Tokuyoshi [2023] gates reuse by fitted vMF similarity (biased, DI only). Salaun et al. [2025] keep the uniform marginal but draw neighbors antithetically from block-local contribution histograms; our renderer's prior spatial reuse implements this and serves as the pair-mixture baseline. Lin et al. [2026] pair pixels reciprocally to halve shift cost but keep random selection. Junkins et al. [2026] importance-sample neighbors by an analytic compatibility score built from G-buffer positions and normals - sample-independent by design so that standard MIS stays valid. SPMIS [Hedstrom et al. 2026] is the closest work and our baseline; we differ in the selection density (learned, shift-aware, target-sensitive at the cell level vs. unshifted source contribution), in the drawing scheme (antithetic stratified vs. i.i.d.), and in gating the neighborhood search.

Learned sampling decisions. Offline, neural nets have selected gradient-domain shift neighbors [Josse et al. 2025; Manzi et al. 2014]. Online-trained tiny networks make per-point sampling decisions in real time for radiance caching [Muller et al. 2021], appearance models [Zeltner et al. 2024], many-light PMFs [Figueiredo et al. 2025], and visibility [Boksansky and Meister 2025]. Our score is deliberately simpler - a reprojected EMA - because the stochastic-MIS compensation removes the usual correctness pressure on the predictor; a fancier model only has to beat the EMA on variance per cost. Sample-dependent decisions that WOULD bias a plain estimator appearing safely inside a compensated

one echoes weighted reservoir resampling theory [Chao 1982; Talbot et al. 2005].

3 Background: GRIS and Stochastic Pairwise MIS

GRIS merges M inputs (X_i, W_i, c_i) into a target pixel by drawing index z proportionally to $w_i = m_i(Y_i) \cdot p(Y_i) \cdot W_i \cdot |\partial T_i / \partial X_i|$, where $Y_i = T_i(X_i)$ is the shifted sample and m_i a resampling MIS weight; the winner's unbiased contribution weight is $W_Y = \Sigma w_i / p(Y)$. Defensive pairwise MIS pairs each non-canonical input with the canonical pixel, at $O(M)$ shifts.

SPMIS replaces m_i by the stochastic $\tilde{m}_i = K(i) / (\tilde{N}P(i)) \cdot m_i$, where a multiset of $\tilde{N}$ indices is drawn with replacement from P , and $K(i)$ is i 's multiplicity. $E[\tilde{m}_i | \text{inputs}] = m_i$ makes the whole GRIS estimate unbiased for any P positive on the support of the contribution; P may depend on the inputs because the expectation is conditioned on them. The canonical weight is estimated separately with $\tilde{N}$ uniform draws ($\tilde{N} = 1$ suffices), and all non-canonical confidences are scaled by $\tilde{N}/M$ to keep the canonical sample from being drowned by stochastic-MIS variance. SPMIS instantiates $P(i) \propto c_i \cdot p(X_i) \cdot W_i$ - the source-pixel contribution - and selects the reuse pool (a screen-space cell of $M \leq 64$ pixels hashed by tile, object and normal) by confidence-weighted WRS with growing search radius.

4 Method

4.1 Factoring the ideal selection weight

The ideal $P(i) \propto m_i(Y_i) \cdot p(Y_i) \cdot W_i \cdot |\partial T_i / \partial X_i|$ factors as

$$P(i) \propto [c_i \cdot p(X_i) \cdot W_i] \cdot \rho_i \cdot \mu_i \quad \rho_i = p(T_i(X_i)) \cdot |\partial T_i / \partial X_i| / p(X_i)$$

where the bracket is SPMIS's source contribution (with c_i in place of the constant), ρ_i is the shift survival ratio, and μ_i collects the remaining MIS-weight variation, which is second order within a reuse pool of similar pixels ($m_i \approx c\Sigma / (c\Sigma + c_i)$) whenever the shift roughly preserves the target). SPMIS sets $\rho_i \mu_i = 1$. Everything our method adds is an estimate of ρ .

ρ is expensive to evaluate on demand - it IS the shift - but a wavefront ReSTIR renderer evaluates thousands of shifts per frame anyway, each yielding one realized ρ sample for one (source, receiver) pair. The signal is already there; it only needs to be remembered in the right place.

4.2 Online shift-survival scores

Each pixel maintains a scalar score v , the exponential moving average ($\alpha = 1/8$, tunable) of bounded survival ratios $\text{clamp}(\rho, 0.02, 1)$ observed on the backward canonical shift the SPMIS chain already runs every frame (the pixel's own sample leaving toward the uniformly drawn partner z'). The 0.02 floor keeps a learned always-fails distinguishable from unlearned (0); writing only one's own score avoids atomics. The observation must be the survival RATIO, not the realized post-shift contribution: the SPMIS weight already carries source brightness, so an absolute observation double-counts it, and its heavy tail (it inherits the unbounded UCW) concentrates P so hard that the $1/(\tilde{N}P)$ compensation sprays fireflies. We measured exactly that failure and its cure (Sec. 5).

The score must track SURFACES, not pixels: under motion we copy it along the temporal reprojection pairing and reset it on disocclusion, exactly mirroring reservoir history. Without this, freshly revealed geometry inherits scores learned on the occluder, under-selects the only contributing sources, and the compensation converts the misprediction into variance spikes.

Selection uses $s_i = \min(c_i, c_{\text{cap}}) \cdot p(X_i) \cdot W_i \cdot (d + (1 - d) \cdot \min(v_i, 1))$ with defensive mix $d = 0.25$ (tunable); unlearned pixels stay neutral. The mix bounds the worst-case weight inflation at $1/d$ and, together with $s_i = 0 \square$ zero contribution, maintains the SPMIS support condition - so unbiasedness holds for ANY score values, learned, stale, or adversarial. Dependence of P on previous frames is covered by conditioning the first-moment argument on the history (Sec. 4.3).

4.3 First-moment lemma and antithetic CDF draws

SPMIS draws the $\tilde{N}$ candidates i.i.d. from P . Inspecting the unbiasedness proof shows only LINEARITY is used: $E[\tilde{m}_i] = E[K(i)] / (\tilde{N}P(i)) \cdot m_i$, so the estimator is unbiased under ANY drawing scheme with $E[K(i)] = \tilde{N}P(i)$, conditioned on the inputs and any auxiliary state P depends on. Draws may be mutually dependent; per-draw marginals matching P suffice. (Lemma and short proof to be spelled out in the final version; it also covers P depending on previous frames' data, since the identity holds pointwise in the conditioning.)

We exploit this twice. First, we draw the k -th of K rounds at the antithetic stratified position $\text{pos}_k = (k \text{ even} ? k/K + u : (k+1)/K - u)$ on the selection CDF, with one Sobol-sequence u per pixel and frame - each pos_k is marginally uniform, so marginals match P exactly while successive draws stratify the selection mass and pair antithetically, composing Salaun-style stratification WITH importance selection (their method is the special case of a uniform target). Second, the learned scores make P depend on all previous frames through the EMA; the conditioned lemma makes that rigorous.

4.4 Starvation-gated cell search

SPMIS always reselects the reuse pool by confidence-weighted WRS over probed cells with growing radius (12 probes from 30 px, $\times 1.25$). On block-run pools that is a structural tax: the own run survives the probe WRS with probability roughly own-mass over total probed mass - about $1/(1+\text{probes})$ at steady state - so most pixels swap close partners for runs up to ~ 350 px away whose shifts fail more. We measure the tax as monotone in BOTH probe count and radius (Sec. 5), and, more surprisingly, that even 64 px/frame fresh disocclusion strips reuse better from their own 256-entry same-frame runs than from distant converged runs: a 16×16 block of fresh RIS candidates is already a viable pool.

The data therefore supports exactly one trigger: a pixel whose run is near-EMPTY (count ≤ 2) has nothing local to reuse and can only gain from foreign runs. This tiny-run gate measures identical to no-search everywhere else while keeping the rescue role; it matters more once cluster keys split by orientation (Sec. 4.5), which multiplies partnerless slivers. Two tempting alternatives fail: gating on mean run confidence breaks under duplication-driven confidence caps

(correlated regions probe forever), and gating on the temporal disocclusion flag hurts the very band it targets. The gate is a sample-value-independent test, so pool choice remains input-set selection needing no MIS compensation; the faithful unconditional search remains available as the baseline configuration.

A third alternative fails for a deeper reason, and the failure is a result. On sparse scenes the natural upgrade is CONTRIBUTION-aware search: trigger when the own run's selection mass S is zero and weight probed runs by THEIR selection mass (confidence is uniform there, so the paper's WRS hops to equally dead runs). Measured, this brightens the sparse scene's converged mean by +16%, concentrated exactly in the dark pixels. The accounting argument is general: the fixed-pool estimator satisfies $E[\text{est}] = P(\text{dead pool}) \cdot 0 + P(\text{live pool}) \cdot E[\text{est} \mid \text{live}] = I$, so realizations with a dead pool MUST contribute zero; a value-dependent pool swap that fires exactly on the dead branch replaces that zero with gathered energy whose full weight the live branches already carry, and no branch-local compensation exists. This is the pool-level form of the value-correlated-selection trap, and it reframes SPMIS's confidence-only search weighting as a CORRECTNESS constraint rather than a missed optimization. The unbiased route is to make foreign runs part of the pool UNCONDITIONALLY - a union pool over value-independently probed runs, drawn through the union's exact selection CDF so the value dependence sits inside the per-draw compensation where the theory allows it, with a deterministic distance kernel keeping the dense-scene tax out. We leave its implementation to future work.

4.5 Implementation

Our renderer's spatial reuse already sorts each jittered 16×16 block's post-temporal reservoirs by (cluster key, target luminance) for histogram stratification; a cluster key is the primary hit's instance, material, and world-normal octant (3 sign bits - the counterpart of SPMIS's objectID + quantized-normal cell hash; adding the normal term measures as a small consistent win on curved geometry). We extend the sort's epilogue with a segmented Hillis-Steele scan producing, per cluster run, the inclusive CDF of the selection weights s_i and the run confidence sum $c\Sigma$ - the contiguous-equal-keys property makes the naive same-key scan exact. Selection inverts the CDF by binary search over ranks, $O(\log M)$ fetches; the drawn candidate's exact probability is the CDF difference over the run mass, with strict comparison making zero-weight ranks unselectable. Per K -round wavefront pass we create one forward shift job (the drawn candidate) plus, in round 0 only, one backward job for the $\tilde{N}_c = 1$ canonical estimate: $K + 1$ shifts per pixel versus the pair-mixture baseline's $2K$ - the stochastic chain is CHEAPER at equal K . The merge folds canonical and candidates into one reservoir chain with the scaled stochastic pairwise weights, maintaining the final-form invariant so temporal reuse and resolve are untouched. The score lives in one persistent vec4 per pixel alongside the disocclusion flag and the chosen pool.

5 Preliminary Results

All numbers below are from the development renderer (RoyalGL, GL compute wavefront ReSTIR BDPT) at 1600×900 on an RTX GPU, Debug host build - RELATIVE comparisons are meaningful, absolute timings are not final. Scenes are Cornell-box variants with one glass duck (plus, where noted, 8 clutter ducks of varied materials); relNoise is the mean absolute 4-neighbor high-frequency residual over the image mean; "masked" restricts it to pixels whose temporal pass found no usable history. The SPARSE-POOL rows use LightMaze, a purpose-built scene (emitter alcove behind a two-bend chicane; a geometric invariant guarantees no straight emitter-to-door line, so NEE is fully occluded and all visible light is ≥ 2 bounces). Confidence cap 8; decorrelation off; normal-octant cluster keys and the tiny-run search gate are the defaults everywhere unless a row says otherwise. All motion tests are FRAME-DETERMINISTIC (fixed per-frame time step, scripted lateral dolly or rocking orbit; stats fire on the frame counter), so cross-configuration comparisons are exactly frame-paired - wall-clock pacing confounds motion A/Bs between configurations of different speed and object-animation tests inherit BLAS-rebuild timing nondeterminism.

Temporal OFF, static, $\tilde{N}$ = 3	pair-mixture	SPMIS [H26]	+ score + lemma draws
glass duck: relNoise	0.192	0.158 (-18%)	0.158
conductor duck: relNoise	0.197	0.162 (-18%)	0.161
rough glass duck: relNoise	0.192	0.158 (-18%)	0.158
spatial pass GPU time	9.1-9.7 ms	7.3-7.6 ms	7.3-7.4 ms

Table 1. Spatial-reuse-only quality (per-frame relNoise, one duck, locked camera). Contribution-proportional selection with stochastic pairwise MIS is the win that survives every test: -18% noise at -20% cost versus the histogram-stratified pair mixture, uniformly across materials. The full mode-2 column (bounded survival score + antithetic stratified CDF positions) matches SPMIS to the third digit: local sorted runs are already ranked well enough by source contribution, and on Cornell-class blocks the selection mass is locally uniform enough that stratifying the CDF draws adds nothing measurable either - the lemma's value here is the license, exercised at zero cost, not a variance win.

Clutter + conductor, temporal ON	SPMIS [H26]	+ survival score	masked (SPMIS / score)
dolly, $\tilde{N} = 3$	0.0984	0.0987	0.458 / 0.461
dolly, $\tilde{N} = 8$	0.0829	0.0831	0.466 / 0.464
orbit, $\tilde{N} = 3$	0.0973	0.0986	0.299 / 0.300
orbit, $\tilde{N} = 8$	0.0762	0.0777	0.274 / 0.276
orbit $\tilde{N}=8$, ABSOLUTE score	-	0.0889, hot 319	0.276
... defensive mix 0.5 / 0.75	-	0.0830 / 0.0791, hot 51 / 17	-
... score factor off	= SPMIS exactly	0.0762, hot 18	0.274

Table 2. Prove-or-kill on the score's home turf: 8 clutter ducks + conductor main duck, frame-deterministic dolly (parallax silhouettes) and 30°/s rocking orbit (64 px/frame fresh edge strips, ~2% of pixels), whole-image / masked per-frame relNoise. The bounded-ratio score is a WASH ($\pm 2\%$, masked included): local runs are already ranked well enough by c-p-W, and $\tilde{N} \in \{3,8\}$ draws cover the contributing set - per-pixel shift survival carries no exploitable extra structure here at any EMA rate we swept. The ABSOLUTE realized-contribution observation (the naive reading of the ideal density) is actively harmful: heavy-tailed scores concentrate P and the $1/(\tilde{N}P)$ compensation inflates reservoir weights into fireflies (hot = pixels brighter than any emitter). The defensive-mixture dose-response (mix 0.25 $\rightarrow$ 0.5 $\rightarrow$ 0.75 gives hot 319 $\rightarrow$ 51 $\rightarrow$ 17 and noise 0.0889 $\rightarrow$ 0.0830 $\rightarrow$ 0.0791 $\rightarrow$ 0.0762 at mix 1) attributes the loss entirely to the score signal; the antithetic draws are free (score-off equals SPMIS to the fourth digit).

Cell search (SPMIS, $\tilde{N}=3$)	dolly whole	orbit whole	orbit masked
no search	0.0894	0.0884	0.298
unconditional, 8 probes / 24 px	0.1098	-	-
unconditional, 8 / 30 px	0.1152	-	-
unconditional, 12 / 24 px	0.1232	-	-
unconditional, 12 / 30 px (paper)	0.1289	0.1289	0.329
tiny-run gate (count ≤ 2)	0.0893	0.0884	0.299
disocclusion-flag gate	0.0894	0.0922	0.328

Table 3. The neighborhood search is a measured steady-state tax on block-run pools: whole-image noise rises monotonically in BOTH probe count and initial radius (frame-paired deterministic dolly), the own run surviving the probe WRS with probability $\sim 1/(1+\text{probes})$. The masked column is the wide fresh strip of a 30°/s orbit - the search's intended beneficiary - where distant converged runs LOSE to the strip's own 256-entry same-frame run, and even a gate that fires exactly there (disocclusion flag) hurts the band it targets. Only near-empty runs benefit from foreign pools: the tiny-run gate equals no-search everywhere measured while keeping that rescue.

LightMaze (sparse pools, temporal OFF)	relNoise K=3	relNoise K=8
no spatial reuse (RIS only)	1.48	1.48
pair-mixture baseline	1.69	1.58
SPMIS selection	0.865 (-49%)	0.519 (-67%)
+ survival score	0.865	0.513
SPMIS + unconditional search	-	1.18
SPMIS + contribution-aware search	-	0.546, mean +16% (BIASED)

Table 4. The sparse-pool regime (LightMaze: emitter behind a two-bend chicane, all visible surfaces ≥ 2 bounces from the light, NEE fully occluded; contributing-entry fraction of reuse runs 1.1% vs 99.7% on Cornell, ~44% of pixels in zero-mass runs; per-frame relNoise, locked camera). Selection inverts from irrelevant to decisive: the pair mixture is WORSE than no reuse (rank-stratified picks feed dead neighbors into the 1/K mixture), SPMIS selection more than halves noise, and in accumulation the SPMIS chain reaches 2.4x lower error at half the samples. The survival score remains a wash even here - finding carriers is the bottleneck, and c-p-W already finds them. The unconditional search stays counterproductive (contribution-blind probes hop to equally dead runs at longer shift distance), and the contribution-aware variant is BIASED per the Sec. 4.4 accounting argument - its +16% lands exactly in the dark pixels. Unbiasedness soaks of the shipped chains hold on this scene (SPMIS spatial-only and temporal+spatial both match the RIS-only mean to $2e-5$; dead runs generate no false-negative events).

Long-average soak (static)	mean luminance
RIS-only / temporal-only / pair-mixture	0.116474 (exact, repeatable 1e-6)
SPMIS chain, $\tilde{N} = 1$	0.116474 (exact)
SPMIS spatial-only, $\tilde{N} = 8$, local runs	0.116509 (+0.030%)
... involutive 48 px partner runs	0.116611 (+0.118%)
... involutive 96 px partner runs	0.116622 (+0.127%)
SPMIS T+S, gated search (default)	0.116550 (+0.065%)
SPMIS T+S, unconditional search	0.116379 (-0.082%)
survival score (mode 2)	= SPMIS band (policy moves no energy)

Table 5. Unbiasedness soaks (locked camera) and the residual's attribution. The stochastic chain carries a small residual whose SIGN follows the reuse regime. Prior bisection rules out selection probabilities, CDF construction, draw accounting, and the estimator structure (a CPU Monte Carlo of the exact chain is unbiased). A dual-direction diagnostic - both shift directions of the SAME (canonical, z') pair each frame, with involutive fixed-stride pairing to keep distant ensembles exchangeable - REFUTES the machinery-asymmetry hypothesis: support-failure classes balance to $1.4e-4$, value ratios to $1e-5$, and the canonical-weight beta channel evaluated through both directions matches to $1e-7$ at local pairs, where the chain is nonetheless $+3.0e-4$ elevated. Instead the residual tracks the direction-SYMMETRIC support-failure rate (1.34%/class local $\rightarrow$ 3.2% at 48 px $\rightarrow$ 3.8% at 96 px, against $+3.0e-4 \rightarrow +11.8e-4 \rightarrow +12.7e-4$): false-negative failures feed the stochastic weight functional asymmetrically in ENERGY - a failed backward shift awards the canonical the full prior share (forcing all backward shifts to fail elevates the mean by 72%, the channel's headroom), while a failed forward shift merely loses one candidate's contribution. Balanced counts, unbalanced magnitudes; the net is positive for local pools, and the unconditional search flips it negative because its role-asymmetric pool choice adds a +7.2% forward-failure excess. The scaling $((c\Sigma - c_c)/(c\Sigma + c_c))^2$ explains $\tilde{N} = 1$ exactness. The fix is therefore not weight symmetrization but raising real shift success (fewer false negatives) - the same lever that dominates variance.

Temporal-free (spatial-only, batched)	static	moving scene	disocc. band	reuse cost
temporal + spatial (K=3, reference)	0.079	0.080	0.415	16-19 ms
spatial-only SPMIS chain, K = 16	0.081	0.079	= whole image	29-35 ms
spatial-only SPMIS chain, K = 24	0.073	-	= whole image	50 ms

Table 6a. Temporal-free ReSTIR via WIDE reuse (per-frame relNoise; batched candidates). Spatial-only reuse at $\tilde{N} = 16$ matches temporal+spatial everywhere while having NO disocclusion band at all (temporal+spatial's history-starved pixels, 1.6% of the frame under this motion, are 5x noisier than its average and it transiently loses ~0.9% energy to history invalidation; spatial-only keeps the reference mean). Iterated spatial passes were tried and REJECTED: re-mixing within-frame aggregates adds almost no sample diversity ($J=2 \times K=3$ is WORSE than one pass of $K=6$) - width plus batching is the right structure, and the missing factor-2 in cost is shift-kernel engineering (reconnection fast path, cheaper light-path replays, reciprocal sharing), not estimator design. Caveat: the stochastic-chain residual of Table 5 grows with $\tilde{N}$ (~+0.07% at 16); its mechanism is now identified (false-negative shift failures), so the same shift-success work that closes the cost gap is expected to shrink it - it must land before this configuration is submission-credible.

Reservoir butterfly (spatial-only)	static	moving scene	spatial cost	soak mean
temporal + spatial (K=3, reference)	0.0794	0.0804	14.5 ms (reuse)	exact
butterfly J = 3 (3 shifts/pixel)	0.0788	0.0748	9.6 ms	0.116476 (exact)
butterfly J = 5	0.0780	-	14.4 ms	0.116477 (exact)
SPMIS chain K = 16 (for scale)	0.0811	0.0794	29-35 ms	+0.07% residual

Table 6b. The RESERVOIR BUTTERFLY: J levels of reciprocal XOR-partner pair merges (partner distance doubling per level, alternating axes, per-frame-jittered grid). Each level costs ONE shift per pixel - the reverse shift IS the partner's forward shift, since XOR pairing is an involution and the cluster test is mutual - and aggregates hold up to 2^J disjoint initial samples: effective samples scale EXPONENTIALLY in shift count. Index-based pairing never inspects sample values (no compensation needed) and the two-way balance-heuristic merge is the symmetric functional: butterfly soaks sit exactly on the unbiased reference, with no stochastic-MIS residual. At three shifts per pixel the spatial-only butterfly matches temporal+spatial statically and clearly beats it under motion, at ~25-35% lower total frame cost, with uniform noise (no disocclusion band), correct energy under motion, and no temporal lag. Deeper butterflies plateau (long-distance shift failure and intra-group correlation, not the confidence cap); J = 3-5 is the sweet spot. Block-scale correlation needs an honest metric beyond high-frequency residuals - duplication maps and accumulated-convergence checks (butterfly J=5 accumulates FASTER than every other configuration) are planned for the final protocol. In our render the butterfly is currently shelved on exactly that ground: the 2^J -pixel groups share aggregates, which reads as a resolution loss up close even where the residual metric improves - the numbers above are reproducible from the archived implementation.

6 Discussion, Limitations, Plan

What is established. The SPMIS reformulation of our spatial reuse is implemented, passes soaks, and is both faster and substantially lower-variance than the histogram-stratified pair-mixture baseline whenever spatial reuse carries real load (-18% noise at -20% cost, Table 1) - with batching, it also

unlocks the temporal-free wide-reuse regime (Table 5a). On top of that baseline, however, SELECTION refinement is saturated on block-local sorted runs: the survival score in its sound bounded-ratio form is a wash at every dose, EMA rate, and scenario we measured (Table 2), stratified CDF draws add nothing on locally-uniform blocks (Table 1), and the naive absolute form of the ideal density is actively harmful through $1/(\tilde{N}P)$ weight inflation - a failure mode we attribute conclusively via its defensive-mixture dose-response. The unconditional neighborhood search is a measured net harm on this scene class at every shape (Table 3); the tiny-run gate keeps its rescue role for free. The chain's small soak residual is attributed by the dual-direction diagnostic: NOT directional machinery asymmetry (bounded at $1e-7$ locally), but direction-symmetric false-negative shift failures whose energy imbalance the stochastic weight functional amplifies (Table 4).

What the findings mean. The regime split is the paper's organizing fact. WHERE a pool holds carriers, variance is dominated by SHIFT OUTCOMES, not by which of 256 same-cluster candidates gets drawn - so survival prediction, stratified draws, and search guidance all wash out, and the right lever is fixing survival itself (deeper-reconnection fallbacks; the same false-negative failures are the identified source of the soak residual). WHERE pools are starved of carriers, plain contribution-proportional selection is decisive (Table 4) - no learned refinement needed, since $c \cdot p \cdot W$ already separates the ~1% of carriers from the dead mass. What remains genuinely open between the regimes is RESCUING zero-mass pools, and the biased contribution-aware search shows this cannot be done by pool-choice heuristics: it requires the union-pool construction with exact per-draw compensation.

What is missing for the paper. (1) The union-pool rescue (Sec. 4.4) - the only identified unbiased mechanism addressing the ~44% zero-mass pixels of the sparse regime. (2) The deeper-reconnection fallback, which serves double duty (variance and residual). (3) The frame-deterministic equal-time protocol on a Release build with MAPE against long references, plus production-scale scenes between the two regime poles (Bistro-class geometry, foliage). (4) A comparison against Junkins-style analytic compatibility as a selection weight, which their and our results suggest should also saturate on dense pools and miss carriers on sparse ones (it is sample-independent by design).

Extensions. Per-target class CDFs (partitioning runs by sub-tile or normal cluster with per-class subtotals) would make P target-dependent with exact probabilities at $O(L)$ extra work; given the saturation result, we expect them to pay off only across strong survival gradients. A tiny-MLP score head is a drop-in EMA replacement under the same unbiasedness guarantee, with the same caveat. The dual-direction diagnostic generalizes to any GRIS system with paired shift directions and turns residual-hunting from bisection into measurement. The method is orthogonal to our volumetric ReSTIR work: fog-cluster runs already flow through the same CDF machinery and a volumetric evaluation is planned as an addition, not a dependency.

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